

Laser-guided bullet

Researchers at Sandia National Laboratories, US, have invented a self-guided bullet for small-caliber, smooth-bore firearms that could hit laser-designated targets about 2000 metres away.

The 10.2cm (4-inch) long bullet includes an optical sensor in the nose to detect the laser beam on a target. The sensed information is sent to guidance and control electronics, which uses an algorithm in an eight-bit central processing unit to command electromagnetic actuators. These actuators steer tiny fins that guide the bullet to the target.

The bullet flies straight due to its aerodynamically stable design, which consists of a centre of gravity that sits forward in the projectile and tiny fins that enable it to fly without spin. Plastic sabots provide a gas seal in the cartridge and protect the delicate fins until these drop off after the bullet emerges from the firearm's barrel.

Double-sided foldable display

Researcher Juergen Steimle is working on double-sided displays that can be folded using predefined hinges. This enables users to dynamically alter both the size and the shape of the display and also access the back side using fold gestures.

The projection-based display is created using six overhead infrared cameras and two high-definition digital projectors that track the movement of the user's head. The image is projected onto a passive white tablet that the user holds. With a number of different designs, Steimle created a range of tablets containing sets of spring-loaded, reversible hinges.

By monitoring the way the tablets are folded, the system uses the act of folding and the resulting form of the tablet as a means of interaction.

Biological computer reading DNA

A group of scientists from The Scripps Research Institute in California and Technion-Israel Institute of Technology has developed a biological computer made entirely from biomolecules, which is capable of deciphering images encrypted on DNA chips.

The hardware and software in these devices are complex biological molecules that activate one another to carry out some predetermined chemical work. The input is a molecule that undergoes specific, predetermined changes following a specific set of rules (software). The output of this chemical computation process is another well-defined molecule.



Scripps Research and Technion logos on a single DNA chip, decrypted into separate fluorescent images using a suitable software

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